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# Golden-Section Search

Derivative-free optimization by interval reduction

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# Four questions

- 1 Why does the constant  $0.618\dots$  appear?
- 2 How can one function value be reused at every step?
- 3 What accuracy is guaranteed after a fixed budget?
- 4 When should we *not* trust the answer?

## The working viewpoint

The method does not chase a derivative. It shrinks a bracket that is known to contain one optimum.

# Problem Setting



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# A derivative-free question

$$x^* = \arg \max_{x \in [x_L, x_U]} f(x)$$

**Only request:** values of  $f(x)$ .

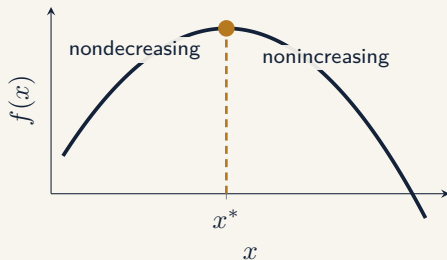
A reliable derivative may be unavailable, expensive, or meaningless.

## Typical situations

- ▶ laboratory measurements;
- ▶ simulation runs;
- ▶ noisy or tabulated output.

For minimization, replace  $\arg \max$  by  $\arg \min$ ; the geometry is unchanged.

# The unimodality contract



A maximum at  $x^*$  divides the bracket:

- ▶  $f$  is nondecreasing up to  $x^*$ ;
- ▶  $f$  is nonincreasing after  $x^*$ .

## What is not required

No derivative, quadratic model, or smooth graph is required.

# The bracket is a claim

- 1 **Valid domain:** every trial point is safe to evaluate.
- 2 **Contains the target:** the desired optimum is inside.
- 3 **One relevant mode:** one peak or trough governs the interval.

## Why this matters

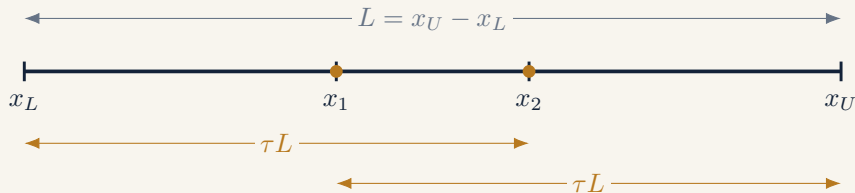
The method certifies a smaller *interval*; it cannot repair a bracket that excluded the optimum or mixed several extrema.

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# The Search Mechanism

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# Two probes, one bracket



$$x_1 = x_U - \tau L$$

$$x_2 = x_L + \tau L$$

$$\tau = \frac{\sqrt{5} - 1}{2} \approx 0.618034, \quad x_L < x_1 < x_2 < x_U.$$

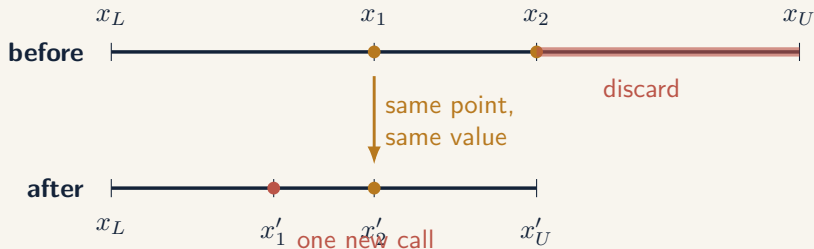
# Which side survives?

| Comparison        | Maximization keeps | Minimization keeps |
|-------------------|--------------------|--------------------|
| $f(x_1) > f(x_2)$ | $[x_L, x_2]$       | $[x_1, x_U]$       |
| $f(x_1) < f(x_2)$ | $[x_1, x_U]$       | $[x_L, x_2]$       |
| $f(x_1) = f(x_2)$ | $[x_1, x_2]$       | $[x_1, x_2]$       |

## Read one row, not the whole table

For a maximum, the larger value points toward the survivor. For a minimum, the smaller value does.

# Reuse one evaluation



For the illustrated maximization case,  $f(x_1) > f(x_2)$ . Therefore

$$x'_U \leftarrow x_2, \quad x'_2 \leftarrow x_1, \quad f(x'_2) \leftarrow f(x_1).$$

Only  $x'_1$  is new, so only **one new function call** is needed.

# Algorithm: initialize once

- 1 Set  $\tau = (\sqrt{5} - 1)/2$  and  $L = x_U - x_L$ .
- 2 Form  $x_1 = x_U - \tau L$  and  $x_2 = x_L + \tau L$ .
- 3 Evaluate  $f_1 = f(x_1)$  and  $f_2 = f(x_2)$ .

## State carried into the loop

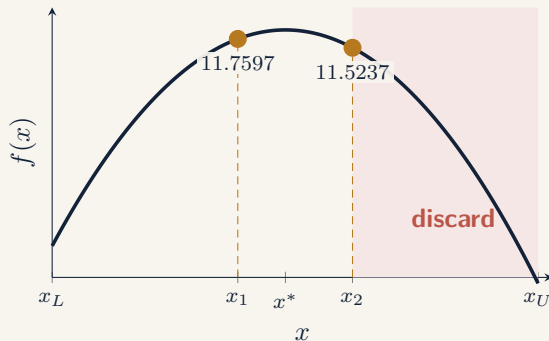
Keep the ordered endpoints, two probes, and their two function values. Losing a saved value also loses the main efficiency advantage.

# Algorithm: repeat

- 1 Compare  $f_1$  and  $f_2$ .
- 2 Discard the side ruled out by unimodality.
- 3 Rename the surviving probe and reuse its value.
- 4 Create one new probe and evaluate  $f$  there.
- 5 Stop when the bracket meets the accuracy rule.

For minimization, reverse only the two strict comparison decisions; the probe geometry and reuse are unchanged.

# Worked trace: first cut



Maximize

$$f(x) = 12 - (x - 2.4)^2$$

on  $[0, 5]$ .

$$x_1 = 1.90983, \quad x_2 = 3.09017.$$

Since

$$f(x_1) > f(x_2),$$

retain  **$[0, 3.09017]$** .

# Worked trace: cuts 0-2

| $k$ | Bracket before       | $x_1$          | $x_2$          | Retained bracket     |
|-----|----------------------|----------------|----------------|----------------------|
| 0   | $[0, 5]$             | 1.90983        | 3.09017        | $[0, 3.09017]$       |
| 1   | $[0, 3.09017]$       | 1.18034        | <b>1.90983</b> | $[1.18034, 3.09017]$ |
| 2   | $[1.18034, 3.09017]$ | <b>1.90983</b> | 2.36068        | $[1.90983, 3.09017]$ |

## Watch the gold entry

It moves to the other probe column because the physical point and its function value are reused.

# Worked trace: cuts 3-5

| $k$ | Bracket before     | $x_1$          | $x_2$          | Retained bracket   |
|-----|--------------------|----------------|----------------|--------------------|
| 3   | [1.90983, 3.09017] | <b>2.36068</b> | 2.63932        | [1.90983, 2.63932] |
| 4   | [1.90983, 2.63932] | 2.18847        | <b>2.36068</b> | [2.18847, 2.63932] |
| 5   | [2.18847, 2.63932] | <b>2.36068</b> | 2.46711        | [2.18847, 2.46711] |

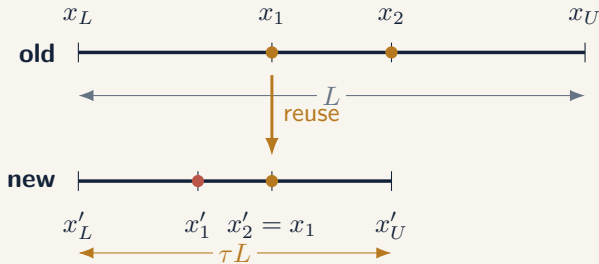
## Result after six comparisons

The true maximizer  $x^* = 2.4$  remains inside  $[2.18847, 2.46711]$ , now only 0.27864 wide.

# Convergence and Cost

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# Why does 0.618 appear?



The old  $x_1$  is  $(1 - \tau)L$  from the left endpoint.

In the new bracket, the right probe is  $\tau(\tau L) = \tau^2 L$  from that endpoint. Reuse requires

$$1 - \tau = \tau^2.$$

Hence

$$\tau = \frac{\sqrt{5} - 1}{2}.$$

# Why the discard is safe

Consider maximization with  $x_1 < x_2$  and  $f(x_1) > f(x_2)$ .

- 1 Suppose, for contradiction, that  $x^* > x_2$ .
- 2 Both probes would then lie on the nondecreasing side of the peak.
- 3 Unimodality would require  $f(x_1) \leq f(x_2)$ , a contradiction.

Therefore  $x^* \leq x_2$ , so discarding  $(x_2, x_U]$  preserves the invariant

$$x^* \in [x_L, x_U].$$

The other comparison and minimization cases are symmetric.

# Rate and error guarantee

## Exact contraction

$$L_k = \tau^k L_0.$$

For midpoint output,

$$|\hat{x}_k - x^*| \leq \frac{L_k}{2}.$$

## Required reductions

$$k \geq \max \left\{ 0, \left\lceil \frac{\log(2\varepsilon/L_0)}{\log \tau} \right\rceil \right\}.$$

## Example

If  $L_0 = 5$  and  $\varepsilon = 10^{-3}$ , then  $k = 17$ . The actual bound is about  $7.00 \times 10^{-4}$ .

# What actually converges?

**Bracket width:** exactly Q-linear,

$$\frac{L_{k+1}}{L_k} = \tau.$$

**Location estimate:** its guaranteed error is at most  $L_k/2$ , also decreasing linearly.

**Objective value:** unimodality alone gives no numerical rate. Extra regularity is needed. For example, if  $f$  is  $M$ -Lipschitz,

$$0 \leq f(x^*) - f(\hat{x}_k) \leq \frac{ML_k}{2}$$

for maximization. A smooth stationary optimum can yield an  $O(L_k^2)$  objective error instead.

# Count calls, not loops

## Initialization

2 function calls

produce the first comparison.

With a stop-after-update implementation,

## Each later comparison

1 new function call

because one value is reused.

$$n \text{ comparisons} \implies n + 1 \text{ evaluations.}$$

For an evaluation budget  $E \geq 2$ , the worst-case bracket is

$$L_{E-1} = \tau^{E-1} L_0.$$

This is why evaluation count is the meaningful complexity measure when  $f$  is a simulation or experiment.

# Fixed-rule alternatives

- 1 **Equal-thirds search:** two new calls per comparison; retain  $2/3$  of the bracket.
- 2 **Fibonacci search:** finite-budget minimax optimal; changing ratios require choosing the budget first.

**Golden section's compromise:** one stationary ratio, one reused value, and no predetermined final budget.

*Reference: Kiefer, "Sequential minimax search for a maximum," Proc. AMS 4 (1953), 502-506.*

# Brent's hybrid

- ▶ Propose an inverse parabolic interpolation step from sampled values.
- ▶ Accept it when safe; otherwise take a golden-section step.

**Trade-off:** golden section gives transparent worst-case contraction. Brent adds bookkeeping to gain speed when the curve is informative.

*Reference: SciPy 1.18 `minimize_scalar` documentation; Brent (1973).*

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# Reliable Computation



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# Stop safely near zero

A common relative test contains  $x_U + x_L$  in the denominator. On a symmetric bracket such as  $[-1, 1]$ , it divides by zero.

For midpoint output, use the scale-aware condition

$$L_k \leq 2 (\varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max\{|x_L|, |x_U|\}).$$

It directly guarantees

$$|\hat{x}_k - x^*| \leq \varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max\{|x_L|, |x_U|\}.$$

Also stop on an evaluation budget or when rounding creates no distinct probe.

# Implementation checks

## Preserve state

- ▶ move each  $(x_i, f_i)$  pair together;
- ▶ cache the surviving evaluation;
- ▶ assert

$$x_L < x_1 < x_2 < x_U.$$

Once machine arithmetic cannot create a distinct probe, further interval algebra cannot create new information.

## Exit honestly

- ▶ reject NaN or Inf values;
- ▶ stop if a probe rounds to an endpoint;
- ▶ return bracket, status, and call count.

# How the method misleads

## Several extrema

A legal-looking comparison can discard the globally best basin.

## Noisy evaluations

One flipped comparison causes an irreversible wrong cut.

## Flat optimum

The bracket shrinks, but the optimizer need not be unique.

## Invalid values

NaN, Inf, or domain errors destroy the ordering decision.

## Wrong inequality

A minimization update used for maximization walks away from the peak.

## False precision

A tiny bracket does not correct model error or measurement bias.

# Modern adaptations

## Line search in many dimensions

$$\phi(\alpha) = F(\mathbf{x} + \alpha \mathbf{d})$$

turns one step-length choice into a scalar golden-section problem.

## Noisy or expensive objectives

- ▶ cache results, repeat uncertain comparisons, and enforce a budget;
- ▶ stop when noise exceeds the improvement being resolved.

The method is sequential: one new call per step saves evaluations, but the next location must wait. Parallel hardware may favor a batched alternative.

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# Mathematical Problems

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# Problem 1: Golden ratio

Normalize a search bracket to  $[0, 1]$ . Place the two probes at

$$x_1 = 1 - \tau, \quad x_2 = \tau, \quad \frac{1}{2} < \tau < 1.$$

Suppose the right-hand bracket  $[x_1, 1]$  survives. Choose  $\tau$  so that the old point  $x_2$  is exactly the left probe of the new bracket.

- 1 Form the self-similarity equation for  $\tau$ .
- 2 Solve it and reject the inadmissible root.
- 3 State the fraction of length removed by one reduction.

No decimal approximation is needed until the last line.

# Solution 1: Derivation

The surviving bracket has lower end  $1 - \tau$  and length  $\tau$ . Its left probe is therefore

$$(1 - \tau) + (1 - \tau)\tau = 1 - \tau^2.$$

Requiring this to coincide with the old point  $x_2 = \tau$  gives

$$1 - \tau^2 = \tau \iff \tau^2 + \tau - 1 = 0.$$

Hence

$$\tau = \frac{-1 \pm \sqrt{5}}{2}.$$

The negative root is outside  $(1/2, 1)$ , so

$$\tau = \frac{\sqrt{5} - 1}{2} \approx 0.618034.$$

The retained fraction is  $\tau$ ; the discarded fraction is  $1 - \tau = \tau^2 \approx 0.381966$ .

# Problem 2: Hand trace

Use golden-section search to minimize

$$f(x) = (x - 1.3)^2 + 2, \quad 0 \leq x \leq 4.$$

With  $\tau = (\sqrt{5} - 1)/2$ , carry out the first **four** bracket reductions. At each step, show

$$x_1 = x_U - \tau(x_U - x_L), \quad x_2 = x_L + \tau(x_U - x_L),$$

the two function values, and the retained bracket.

Finally, report the midpoint estimate and the number of function evaluations used.

# Solution 2: Trace table

For minimization,  $f(x_1) < f(x_2)$  keeps  $[x_L, x_2]$ ; otherwise it keeps  $[x_1, x_U]$ .

| $k$ | $x_1$  | $x_2$  | $f(x_1)$ | $f(x_2)$ | retained bracket   |
|-----|--------|--------|----------|----------|--------------------|
| 1   | 1.5279 | 2.4721 | 2.0519   | 3.3739   | $[0, 2.4721]$      |
| 2   | 0.9443 | 1.5279 | 2.1265   | 2.0519   | $[0.9443, 2.4721]$ |
| 3   | 1.5279 | 1.8885 | 2.0519   | 2.3464   | $[0.9443, 1.8885]$ |
| 4   | 1.3050 | 1.5279 | 2.0000   | 2.0519   | $[0.9443, 1.5279]$ |

Thus

$$\hat{x} = \frac{0.9443 + 1.5279}{2} = \boxed{1.2361}.$$

Two calls initialize the method; each later reduction needs one new call. Four reductions therefore use  $2 + 3 = \boxed{5}$  evaluations.

Check: the exact minimizer 1.3 remains inside every retained bracket.

# Problem 3: Error budget

A unimodal objective is bracketed on  $[2, 18]$ . After  $k$  golden-section reductions, the midpoint of the surviving bracket will be reported.

Find the smallest integer  $k$  that guarantees

$$|\hat{x} - x^*| \leq 5 \times 10^{-4}.$$

Your solution must:

- 1 start from the bracket-length formula;
- 2 verify that one fewer reduction is insufficient;
- 3 convert the result into a function-evaluation budget.

# Solution 3: Budget

The initial length is  $L_0 = 18 - 2 = 16$ , and after  $k$  reductions

$$L_k = 16\tau^k, \quad |\hat{x} - x^*| \leq \frac{L_k}{2} = 8\tau^k.$$

Therefore

$$8\tau^k \leq 5 \times 10^{-4} \quad \implies \quad k \geq \frac{\ln(6.25 \times 10^{-5})}{\ln \tau} \approx 20.12.$$

Thus the smallest integer is  $\boxed{k = 21}$ .

The boundary check is

$$8\tau^{20} \approx 5.289 \times 10^{-4} > 5 \times 10^{-4}, \quad 8\tau^{21} \approx 3.269 \times 10^{-4} < 5 \times 10^{-4}.$$

With cached reuse,  $k$  reductions require  $k + 1$  calls. The guaranteed budget is therefore  $\boxed{22 \text{ function evaluations}}$ .

# Problem 4: Safe discard

Let  $f$  be strictly unimodal with a unique maximum  $x^*$  on  $[x_L, x_U]$ , and let

$$x_L < x_1 < x_2 < x_U.$$

Prove directly from strict unimodality that

$$f(x_1) < f(x_2) \implies x^* \in [x_1, x_U].$$

Then determine what the equality  $f(x_1) = f(x_2)$  says about the location of  $x^*$ .

## Allowed fact

A strictly unimodal function increases before  $x^*$  and decreases after  $x^*$ .

# Solution 4: Proof

**Case 1:**  $f(x_1) < f(x_2)$ .

Suppose instead that  $x^* < x_1$ . Both probes would then lie on the strictly decreasing side, so

$$x^* < x_1 < x_2 \implies f(x_1) > f(x_2),$$

a contradiction. Therefore

$$x^* \in [x_1, x_U].$$

**Case 2:**  $f(x_1) = f(x_2)$ .

If  $x^* \leq x_1$ , strict decrease gives  $f(x_1) > f(x_2)$ . If  $x^* \geq x_2$ , strict increase gives  $f(x_1) < f(x_2)$ . Both are impossible, hence

$$x^* \in (x_1, x_2).$$

A flat-topped objective needs a weaker conclusion about its set of maximizers.

# Problem 5: Stopping rule

Consider the relative-width test

$$2 \left| \frac{x_U - x_L}{x_U + x_L} \right| < \delta.$$

It is applied to  $[x_L, x_U] = [-0.002, 0.002]$  with  $\delta = 10^{-3}$ .

- 1 Explain exactly why the test fails.
- 2 Replace it with a mixed absolute-relative test for the midpoint, using  $\varepsilon_{\text{abs}} = 10^{-4}$  and  $\varepsilon_{\text{rel}} = 10^{-3}$ .
- 3 Find how many additional golden reductions guarantee the new test.

# Solution 5: Scale floor

Here  $x_U + x_L = 0$ , so the original test is undefined even though the bracket length is 0.004. Use

$$\frac{L}{2} \leq \varepsilon_{\text{abs}} + \varepsilon_{\text{rel}} \max(|x_L|, |x_U|).$$

The present length tolerance is  $2.04 \times 10^{-4}$ . Since the relative scale can fall as the bracket moves, a uniform guarantee uses the absolute floor  $L \leq 2 \times 10^{-4}$ . With  $L_m = 0.004\tau^m$ ,

$$m \geq \frac{\ln(2 \times 10^{-4}/0.004)}{\ln \tau} \approx 6.23, \quad \boxed{m = 7}.$$

Indeed,  $L_6 \approx 2.229 \times 10^{-4}$  fails, while  $L_7 \approx 1.378 \times 10^{-4}$  passes. The absolute term prevents the denominator pathology near zero.

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# Application Problems

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# Application 1: Damper

A suspension engineer tunes a dimensionless damping setting  $c$ . A finite-element run predicts the peak displacement

$$D(c) = 1.5 + 12(c - 0.38)^2 \text{ mm}, \quad 0.10 \leq c \leq 0.90.$$

The engineer treats the simulator as a black box and minimizes  $D$  using golden-section search. One run takes 8 minutes.

- 1 Carry out the first three bracket reductions.
- 2 Report the final bracket and its midpoint.
- 3 Find the total simulation time when values are cached.

The formula is supplied so that the simulator outputs can be checked by hand.

# Solution 1: Damper result

For minimization, retain the side containing the smaller displacement.

| $k$ | $c_1$  | $c_2$  | $D(c_1)$ | $D(c_2)$ | retained bracket   |
|-----|--------|--------|----------|----------|--------------------|
| 1   | 0.4056 | 0.5944 | 1.5078   | 2.0517   | $[0.1000, 0.5944]$ |
| 2   | 0.2889 | 0.4056 | 1.5997   | 1.5078   | $[0.2889, 0.5944]$ |
| 3   | 0.4056 | 0.4777 | 1.5078   | 1.6146   | $[0.2889, 0.4777]$ |

Hence

$$c \in [0.2889, 0.4777], \quad \hat{c} = \frac{0.2889 + 0.4777}{2} = \boxed{0.3833}.$$

Three reductions need  $3 + 1 = 4$  cached evaluations, so the elapsed simulation time is  $4(8) = \boxed{32 \text{ minutes}}$ .

Audit: the model's exact best setting,  $c = 0.38$ , is inside the bracket.

# Application 2: Insulation

A building model balances insulation cost against heat loss:

$$C(t) = 180t + 100e^{-10t}, \quad 0.02 \leq t \leq 0.32,$$

where  $t$  is thickness in metres and  $C$  is annual cost in thousands of taka. One energy-model run takes 45 seconds.

Use golden-section search to guarantee a midpoint thickness error of at most 1 mm.

- 1 Determine the required reductions and model calls.
- 2 Convert the call count into elapsed time.
- 3 Report the computed bracket and midpoint thickness.

# Solution 2: Result

Here  $L_0 = 0.32 - 0.02 = 0.30$  m. The midpoint guarantee requires

$$\frac{0.30\tau^k}{2} \leq 0.001 \implies k \geq \frac{\ln(0.001/0.15)}{\ln \tau} \approx 10.41.$$

Thus  $k = 11$  reductions and 12 model calls are needed. The corresponding time is

$$12(45) = 540 \text{ s} = 9 \text{ minutes}.$$

Running the updates gives

$$[t_L, t_U] = [0.170466, 0.171973] \text{ m}, \quad \hat{t} = 0.171220 \text{ m} = 171.2 \text{ mm}.$$

Independent audit:  $C'(t) = 0$  gives  $t^* = \ln(1000/180)/10 = 0.171480$  m, which lies in the bracket.

# Application 3: Dosage

A dose-response surrogate assigns the net clinical benefit

$$B(d) = 95 - 0.12(d - 14)^2, \quad 5 \leq d \leq 25,$$

where  $d$  is a dose in milligrams. The approved interval is unimodal, and the benefit must be maximized without using derivatives.

The design requirement is stated in outcome units:

$$B(d^*) - B(\hat{d}) \leq 0.05 \text{ benefit points.}$$

Find the smallest guaranteed reduction count, the evaluation count, and the midpoint recommendation produced by golden-section search.

# Solution 3: Benefit bound

After  $k$  reductions, the midpoint error obeys

$$|\hat{d} - d^*| \leq 10\tau^k \implies B(d^*) - B(\hat{d}) \leq 0.12(10\tau^k)^2.$$

Requiring this bound to be at most 0.05 gives

$$0.12(10\tau^k)^2 \leq 0.05 \implies k \geq 5.69.$$

Therefore  $\boxed{k = 6}$ , requiring  $\boxed{7}$  benefit evaluations.

The boundary check is  $0.0976 > 0.05$  at  $k = 5$ , but  $0.0373 < 0.05$  at  $k = 6$ . Six updates give  $[13.3282, 14.4427]$  mg, so

$$\boxed{\hat{d} = 13.8854 \text{ mg}}.$$

Its actual model loss is only 0.0016; 0.0373 is the advance guarantee.

# Application 4: Pricing

A retailer tests price  $p$ , measured in hundreds of taka. A demand simulator predicts weekly profit, in thousands of taka, as

$$\Pi(p) = 12(p - 8)e^{-0.06p}, \quad 10 \leq p \leq 40.$$

Each simulator call takes 20 minutes.

- 1 Compute the initial golden probes and their profits.
- 2 Perform the first maximization update and identify the reused point.
- 3 For eight reductions, compare cached golden search with an implementation that evaluates two probes every time.

# Solution 4: Reuse pays

The initial probes and profits are

$$p_1 = 21.4590, \quad \Pi(p_1) = 44.568, \quad p_2 = 28.5410, \quad \Pi(p_2) = 44.472.$$

Since  $\Pi(p_1) > \Pi(p_2)$ , maximization retains

$$\boxed{[10, 28.5410]}.$$

The old  $p_1 = 21.4590$  becomes the new right probe; only the new left probe

$$p'_1 = 28.5410 - \tau(28.5410 - 10) = 17.0820$$

needs evaluation.

For eight reductions,

$$\text{cached calls} = 8 + 1 = 9, \quad \text{two fresh probes each time} = 16.$$

Reuse saves 7 calls, or  $7(20) = \boxed{140 \text{ minutes}}.$

# Application 5: Solar tilt

A solar-field simulation maximizes daily energy over  $10^\circ \leq \theta \leq 50^\circ$ . The first golden probes are

$$\theta_1 = 25.279^\circ, \quad \theta_2 = 34.721^\circ.$$

Weather variability produces these 95% intervals, in kWh:

$$\theta_1 : 5.84 \pm 0.06, \quad \theta_2 : 5.88 \pm 0.06.$$

After replication they narrow to  $[5.82, 5.86]$  and  $[5.87, 5.91]$ .

Explain the correct decision before and after replication, find the next new probe, and calculate the cost of five runs per probe at two minutes per run.

# Solution 5

Initially the intervals overlap, so the sign of  $E(\theta_2) - E(\theta_1)$  is unresolved. A standard golden-section discard would risk deleting the true maximum.

After replication,

$$[5.82, 5.86] \cap [5.87, 5.91] = \emptyset,$$

and the second probe is reliably better. For maximization, retain

$$\boxed{[25.279^\circ, 50^\circ]}.$$

The old  $34.721^\circ$  point becomes the new left probe. The new right probe is

$$25.279 + \tau(50 - 25.279) = \boxed{40.557^\circ}.$$

Five runs at each original probe require 10 runs, costing  $10(2) = \boxed{20 \text{ minutes}}$ . Later comparisons should also replicate only when their uncertainty intervals overlap.

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# Summary & References

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# Project repository

**LaTeX source, figures, and reproducibility files**

`github.com/JonayedMohiuddin/golden-section-search`

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